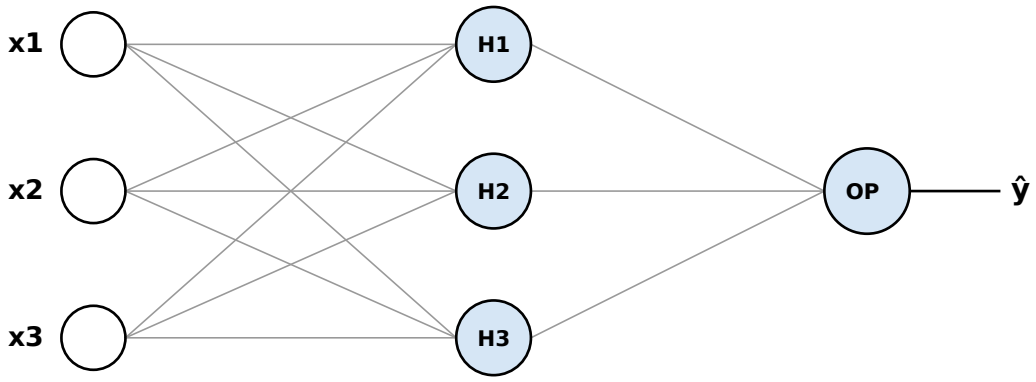
MLP Coursework – Deep Learning Module

Problem 1 – Solution



(3 inputs, 3 hidden nodes, 1 output node)

1. Weights in the network

The network has 3 inputs (x_1, x_2, x_3), 3 hidden nodes (H1, H2, H3) and 1 output node (OP). W_{ij} is the weight from input i to hidden node j, and W_{0j} is the bias of hidden node j.

Node	Weights	Bias
H1	W_{11}, W_{21}, W_{31}	W_{01}
H2	W_{12}, W_{22}, W_{32}	W_{02}
H3	W_{13}, W_{23}, W_{33}	W_{03}
OP	W_{1op}, W_{2op}, W_{3op}	W_{0op}

2. Number of parameters

Input → hidden: 3 inputs × 3 hidden nodes = 9 weights, plus 3 biases = 12

Hidden → output: 3 weights + 1 bias = 4

Total parameters = 12 + 4 = 16

3. Forward transformation equations

Using sigmoid, $\sigma(z) = 1 / (1 + e^{-z})$.

Hidden layer (z is the weighted sum, h is the output after activation):

$$z_1 = W_{01} + W_{11}x_1 + W_{21}x_2 + W_{31}x_3, \quad h_1 = \sigma(z_1)$$
$$z_2 = W_{02} + W_{12}x_1 + W_{22}x_2 + W_{32}x_3, \quad h_2 = \sigma(z_2)$$
$$z_3 = W_{03} + W_{13}x_1 + W_{23}x_2 + W_{33}x_3, \quad h_3 = \sigma(z_3)$$

Output layer:

$$z_{op} = W_{0op} + W_{1op}h_1 + W_{2op}h_2 + W_{3op}h_3$$
$$\hat{y} = \sigma(z_{op})$$

4. Gradients (MSE loss)

Loss for one sample: $L = \frac{1}{2}(y - \hat{y})^2$. (The ½ is only there to cancel the 2 while differentiating.)

Useful derivative: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, so $d\hat{y}/dz_{op} = \hat{y}(1 - \hat{y})$ and $dh_j/dz_j = h_j(1 - h_j)$.

(a) Output node weights

By chain rule, $\partial L/\partial W_{iop} = \partial L/\partial \hat{y} \times \partial \hat{y}/\partial z_{op} \times \partial z_{op}/\partial W_{iop}$

$\partial L/\partial \hat{y} = -(y - \hat{y}) = (\hat{y} - y)$

$\partial \hat{y}/\partial z_{op} = \hat{y}(1 - \hat{y})$

$\partial z_{op}/\partial W_{iop} = h_i$ (and 1 for the bias W_{0op})

Let $\delta_{op} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y})$. Then:

$$\boldsymbol{\partial L/\partial W_{1op} = \delta_{op} \cdot h_1}$$
$$\boldsymbol{\partial L/\partial W_{2op} = \delta_{op} \cdot h_2}$$
$$\boldsymbol{\partial L/\partial W_{3op} = \delta_{op} \cdot h_3}$$
$$\boldsymbol{\partial L/\partial W_{0op} = \delta_{op}}$$

(b) H3 weights (W_{13}, W_{23}, W_{33} and bias W_{03})

The error reaches H3 only through the output node, so the chain is longer:

$\partial L/\partial W_{i3} = \partial L/\partial z_{op} \times \partial z_{op}/\partial h_3 \times \partial h_3/\partial z_3 \times \partial z_3/\partial W_{i3}$

$\partial L/\partial z_{op} = \delta_{op} = (\hat{y} - y)\ \hat{y}(1 - \hat{y})$

$\partial z_{op}/\partial h_3 = W_{3op}$

$\partial h_3/\partial z_3 = h_3(1 - h_3)$

$\partial z_3/\partial W_{i3} = x_i$ (and 1 for the bias)

Let $\delta_3 = \delta_{op} \cdot W_{3op} \cdot h_3(1 - h_3)$. Then:

$$\boldsymbol{\partial L/\partial W_{13} = \delta_3 \cdot x_1}$$
$$\boldsymbol{\partial L/\partial W_{23} = \delta_3 \cdot x_2}$$
$$\boldsymbol{\partial L/\partial W_{33} = \delta_3 \cdot x_3}$$
$$\boldsymbol{\partial L/\partial W_{03} = \delta_3}$$

Fully expanded: $\partial L/\partial W_{i3} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) \cdot W_{3op} \cdot h_3(1 - h_3) \cdot x_i$

These are used in gradient descent as $W \leftarrow W - \eta \cdot \partial L/\partial W$, where η is the learning rate.